ANSWER KEY & MARKING SCHEME SET 1

Chapter 6: **We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** 1287	$99 \times 13 = (100 - 1) \times 13$
2	**(b)** 91	$7 \times 10 + 7 \times 3$
3	**(c)** $3x + 12$	Multiply each term inside by the outside number
4	**(d)** $a^2 + 2ab + b^2$	Standard identity
5	**(a)** 441	$21^2 = 21 \times 21$
6	**(a)** 17	$3 \times 4 + 5$
7	**(a)** 7	The number multiplying x
8	**(d)** 2424	$101 \times 24 = (100 + 1) \times 24$
9	**(d)** 324	$6 \times 50 + 6 \times 4$
10	**(c)** $5x + 10$	Multiply each term inside by the outside number

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$99 \times 16 = (100 - 1) \times 16$ (1) $= 1600 - 16 = 1584$ (1).	1+1
12	$7 \times x + 7 \times 3$ (1) $= 7x + 21$ (1).	1+1
13	$34 = 30 + 4$ (1); $30^2 + 2 \times 30 \times 4 + 4^2 = 1156$ (1).	1+1
14	$102 \times 25 = (100 + 2) \times 25$ (1) $= 2500 + 50 = 2550$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$2x + 6 + 4x + 20$ (1); collect like terms $= 6x + 26$ (2).	1+2
16	$x^2 + 3x + 2x + 6$ (2); $= x^2 + 5x + 6$ (1).	2+1
17	$24 = 20 + 4$ (1); $20^2 + 2(20)(4) + 4^2 = 400 + 160 + 16$ (1); $= 576$ (1).	1+1+1
	OR: $3x + 6 + 5x + 5$ (1); collect like terms $= 8x + 11$ (2).	1+2
18	$x^2 + 5x + 4x + 20$ (2); $= x^2 + 9x + 20$ (1).	2+1
	OR: $36 = 30 + 6$ (1); $30^2 + 2(30)(6) + 6^2 = 900 + 360 + 36$ (1); $= 1296$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$x^2 + 4x + 2x + 8$ (2); $= x^2 + 6x + 8$ (1); at $x = 3$: $9 + 18 + 8 = 35$ (2).	2+1+2
	OR: $104 \times 25 = (100 + 4) \times 25 = 2500 + 100$ (2); $= 2600$ (1); direct multiplication gives 2600 (2).	2+1+2
20	$x^2 + 5x + 3x + 15$ (2); $= x^2 + 8x + 15$ (1); at $x = 2$: $4 + 16 + 15 = 35$ (2).	2+1+2
	OR: $96 \times 35 = (100 - 4) \times 35 = 3500 - 140$ (2); $= 3360$ (1); direct multiplication gives 3360 (2).	2+1+2

Mistakes to avoid:

- Writing $(a + b)^2 = a^2 + b^2$; the middle term $2ab$ is missed.
- Writing $(a - b)^2 = a^2 - b^2$; correct is $a^2 - 2ab + b^2$, while $a^2 - b^2 = (a + b)(a - b)$.
- Multiplying only the first term of a bracket: $3(x + 4) = 3x + 4$ instead of $3x + 12$.
- Sign errors when multiplying: $-2 \times (-3x) = +6x$ and $(x - 3)(x - 2)$ has $+6$ as the constant term.
- Adding unlike terms: $2x + 3x^2 = 5x^3$; x and x^2 are not like terms.
- Squaring a product without squaring the coefficient: $(3x)^2 = 9x^2$, not $3x^2$.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

ANSWER KEY & MARKING SCHEME SET 2

Chapter 6: **We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** $a^2 - 2ab + b^2$	Standard identity
2	**(d)** 1024	$32^2 = 32 \times 32$
3	**(c)** 19	$2 \times 6 + 7$
4	**(a)** 4	The number multiplying x
5	**(c)** 1470	$98 \times 15 = (100 - 2) \times 15$
6	**(a)** 200	$8 \times 20 + 8 \times 5$
7	**(b)** $4x + 24$	Multiply each term inside by the outside number
8	**(d)** $a^2 - b^2$	Standard identity
9	**(b)** 1681	$41^2 = 41 \times 41$
10	**(c)** 16	$5 \times 3 + 1$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$4 \times x + 4 \times 8$ (1) $= 4x + 32$ (1).	1+1
12	$52 = 50 + 2$ (1); $50^2 + 2 \times 50 \times 2 + 2^2 = 2704$ (1).	1+1
13	$98 \times 22 = (100 - 2) \times 22$ (1) $= 2200 - 44 = 2156$ (1).	1+1
14	$9 \times x + 9 \times 5$ (1) $= 9x + 45$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$4x + 4 + 2x + 12$ (1); collect like terms $= 6x + 16$ (2).	1+2
16	$x^2 + 6x + 8x + 48$ (2); $= x^2 + 14x + 48$ (1).	2+1
17	$43 = 40 + 3$ (1); $40^2 + 2(40)(3) + 3^2 = 1600 + 240 + 9$ (1); $= 1849$ (1).	1+1+1
	OR: $5x + 20 + 3x + 6$ (1); collect like terms $= 8x + 26$ (2).	1+2
18	$x^2 + 7x + 3x + 21$ (2); $= x^2 + 10x + 21$ (1).	2+1
	OR: $58 = 50 + 8$ (1); $50^2 + 2(50)(8) + 8^2 = 2500 + 800 + 64$ (1); $= 3364$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$x^2 + 6x + 8x + 48$ (2); $= x^2 + 14x + 48$ (1); at $x = 4$: $16 + 56 + 48 = 120$ (2).	2+1+2
	OR: $105 \times 18 = (100 + 5) \times 18 = 1800 + 90$ (2); $= 1890$ (1); direct multiplication gives 1890 (2).	2+1+2
20	$x^2 + 7x + 4x + 28$ (2); $= x^2 + 11x + 28$ (1); at $x = 1$: $1 + 11 + 28 = 40$ (2).	2+1+2
	OR: $95 \times 24 = (100 - 5) \times 24 = 2400 - 120$ (2); $= 2280$ (1); direct multiplication gives 2280 (2).	2+1+2

Mistakes to avoid:

- Writing $(a + b)^2 = a^2 + b^2$; the middle term $2ab$ is missed.
- Writing $(a - b)^2 = a^2 - b^2$; correct is $a^2 - 2ab + b^2$, while $a^2 - b^2 = (a + b)(a - b)$.
- Multiplying only the first term of a bracket: $3(x + 4) = 3x + 4$ instead of $3x + 12$.
- Sign errors when multiplying: $-2 \times (-3x) = +6x$ and $(x - 3)(x - 2)$ has $+6$ as the constant term.
- Adding unlike terms: $2x + 3x^2 = 5x^3$; x and x^2 are not like terms.
- Squaring a product without squaring the coefficient: $(3x)^2 = 9x^2$, not $3x^2$.

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CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 3

Chapter 6: **We Distribute, Yet Things Multiply** | Class 8 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** 8	The number multiplying x
2	**(b)** 1836	$102 \times 18 = (100 + 2) \times 18$
3	**(b)** 288	$9 \times 30 + 9 \times 2$
4	**(b)** $2x + 18$	Multiply each term inside by the outside number
5	**(b)** 2809	$53^2 = 53 \times 53$
6	**(a)** 29	$4 \times 5 + 9$
7	**(b)** 3	The number multiplying x
8	**(b)** 1940	$97 \times 20 = (100 - 3) \times 20$
9	**(c)** 235	$5 \times 40 + 5 \times 7$
10	**(d)** $6x + 18$	Multiply each term inside by the outside number

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$63 = 60 + 3$ (1); $60^2 + 2 \times 60 \times 3 + 3^2 = 3969$ (1).	1+1
12	$104 \times 15 = (100 + 4) \times 15$ (1) $= 1500 + 60 = 1560$ (1).	1+1
13	$6 \times x + 6 \times 7$ (1) $= 6x + 42$ (1).	1+1
14	$47 = 40 + 7$ (1); $40^2 + 2 \times 40 \times 7 + 7^2 = 2209$ (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$6x + 12 + 7x + 49$ (1); collect like terms $= 13x + 61$ (2).	1+2
16	$x^2 + 9x + 2x + 18$ (2); $= x^2 + 11x + 18$ (1).	2+1
17	$62 = 60 + 2$ (1); $60^2 + 2(60)(2) + 2^2 = 3600 + 240 + 4$ (1); $= 3844$ (1).	1+1+1
	OR: $2x + 16 + 3x + 9$ (1); collect like terms $= 5x + 25$ (2).	1+2
18	$x^2 + 6x + 5x + 30$ (2); $= x^2 + 11x + 30$ (1).	2+1
	OR: $71 = 70 + 1$ (1); $70^2 + 2(70)(1) + 1^2 = 4900 + 140 + 1$ (1); $= 5041$ (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$x^2 + 5x + 2x + 10$ (2); $= x^2 + 7x + 10$ (1); at $x = 5$: $25 + 35 + 10 = 70$ (2).	2+1+2
	OR: $108 \times 15 = (100 + 8) \times 15 = 1500 + 120$ (2); $= 1620$ (1); direct multiplication gives 1620 (2).	2+1+2
20	$x^2 + 6x + 3x + 18$ (2); $= x^2 + 9x + 18$ (1); at $x = 3$: $9 + 27 + 18 = 54$ (2).	2+1+2
	OR: $92 \times 30 = (100 - 8) \times 30 = 3000 - 240$ (2); $= 2760$ (1); direct multiplication gives 2760 (2).	2+1+2

Mistakes to avoid:

- Writing $(a + b)^2 = a^2 + b^2$; the middle term $2ab$ is missed.
- Writing $(a - b)^2 = a^2 - b^2$; correct is $a^2 - 2ab + b^2$, while $a^2 - b^2 = (a + b)(a - b)$.
- Multiplying only the first term of a bracket: $3(x + 4) = 3x + 4$ instead of $3x + 12$.
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